

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-acba873da91502641.jsonl`  
- SHA-256: `aa4fba68704e53bf65a85c21e4faa494bcb2e569c61b36b47b89b2ffae719085`  
- Source modified: `2026-06-16T03:24:52+00:00`  
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- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:24:15.452Z)

Adversarial Claim Verifier (voter 3/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research question: ****What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?**** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best ***recent*** methods for (a) turning a dense per-frame signal into well-segmented

actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k, edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible.

Claim under review

"Detecting action boundaries with a dedicated branch yields large gains specifically on over-segmentation-sensitive metrics: up to 13.7% improvement in segmental edit distance and up to 16.1% in segmental F1 score over prior state-of-the-art."

****Source:**** <https://arxiv.org/abs/2007.06866> (primary)

****Supporting quote:**** "improvements of "up to 13.7% in terms of segmental edit distance and up to 16.1% in terms of segmental F1 score" over prior state-of-the-art methods"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-16T03:24:19.195Z)

I'll verify this claim about action boundary regression yielding gains on over-segmentation metrics. The source arXiv ID (2007.06866) and the specific numbers (13.7% edit distance, 16.1%

segmental F1) point to ASRF (Alleviating Over-segmentation Errors by Detecting Action Boundaries). Let me check.

3. assistant (2026-06-16T03:24:22.064Z)

Let me fetch the arXiv abstract directly to verify the numbers.

4. user (2026-06-16T03:24:27.191Z)

Paper Details

Title: Alleviating Over-segmentation Errors by Detecting Action Boundaries

Authors: Yuchi Ishikawa, Seito Kasai, Yoshimitsu Aoki, and Hirokatsu Kataoka

Improvement Numbers: The abstract reports improvements of up to 13.7% in segmental edit distance and up to 16.1% in segmental F1 score.

Verbatim Quote: "Our framework outperforms state-of-the-art methods on three challenging datasets, offering an improvement of up to 13.7% in terms of segmental edit distance and up to 16.1% in terms of segmental F1 score."

Confirmation: Yes, this is the ASRF (Action Segment Refinement Framework) paper, as confirmed by the abstract which describes the framework's two-branch architecture for temporal action segmentation.

5. user (2026-06-16T03:24:31.978Z)

Web search results for query: "ASRF "Alleviating Over-segmentation Errors by Detecting Action Boundaries" "13.7%" edit distance "16.1%" segmental F1"

Links: [{"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries | IEEE Conference Publication | IEEE Xplore", "url": "https://ieeexplore.ieee.org/document/9423184/"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries", "url": "https://yiskw713.github.io/asrf/"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries | DeepAI", "url": "https://deepai.org/publication/alleviating-over-segmentation-errors-by-detecting-action-boundaries"}, {"title": "asrf/evaluate.py at main · yiskw713/asrf", "url": "https://github.com/yiskw713/asrf/blob/main/evaluate.py"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries", "url": "https://arxiv.org/pdf/2007.06866"}, {"title": "GitHub - yiskw713/asrf: Code for 'Alleviating Over-segmentation Errors by Detecting Action Boundaries' accepted in WACV2021", "url": "https://github.com/yiskw713/asrf"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries", "url": "https://openaccess.thecvf.com/content/WACV2021/papers/Ishikawa_Alleviating_Over-Segmentation_Errors_by_Detecting_Action_Boundaries_WACV_2021_paper.pdf"}, {"title": "WACV 2021 Open Access Repository", "url": "https://openaccess.thecvf.com/content/WACV2021/html/Ishikawa_Alleviating_Over-Segmentation_Errors_by_Detecting_Action_Boundaries_WACV_2021_paper.html"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries | Request PDF", "url": "https://www.researchgate.net/publication/342944719_Alleviating_Over-segmentation_Errors_by_Detecting_Action_Boundaries"}]

I found the information you were looking for! The search results contain details about ASRF and the specific performance metrics you mentioned.

ASRF (Action Segment Refinement Framework) is a framework for the temporal action segmentation task that consists of a long-term feature extractor and two branches: the Action Segmentation Branch (ASB) and the Boundary Regression Branch (BRB).

The model outperforms state-of-the-art methods on three challenging datasets, offering an improvement of up to 13.7% in terms of segmental edit distance and up to 16.1% in terms of segmental F1 score.

The paper was accepted to WACV 2021.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-16T03:24:43.164Z)

The claim is directly verified against the primary source. Both the WebFetch of the arXiv abstract and an independent WebSearch (which surfaced the IEEE Xplore, CVF WACV 2021 open-access PDF, and the authors' project page) confirm the exact figures and framing.

Let me note the one nuance worth checking for accuracy of the claim's phrasing: the claim says boundaries "yield large gains specifically on over-segmentation-sensitive metrics." The paper's headline numbers are indeed reported on edit distance and segmental F1 ? both of which are the standard over-segmentation-sensitive metrics. The claim is faithful to the source.

Checklist verdict:

1. ****Supported by quote?**** Yes ? verbatim match. Abstract: "an improvement of up to 13.7% in terms of segmental edit distance and up to 16.1% in terms of segmental F1 score." The claim's "up to" qualifier is preserved (not overstated as a uniform/mean gain).
2. ****Contradicting evidence?**** None found. Independent sources (IEEE Xplore, CVF, project page) corroborate identical numbers.
3. ****Source quality?**** Peer-reviewed (WACV 2021), primary source. Sufficient for the claim's strength.
4. ****Outdated?**** 2020/2021. The numbers are a historical, correctly-attributed result ("over prior state-of-the-art" = the SOTA at that time, primarily MS-TCN/MS-TCN++). The claim does not assert these are current SOTA, so the date does not undermine it.
5. ****Marketing/cherry-pick?**** No ? peer-reviewed result with public code (github.com/yiskw713/asrf) and standard benchmarks (50Salads, GTEA, Breakfast).

One minor caveat for honesty: the framing "boundaries yield large gains specifically on over-segmentation-sensitive metrics" is the claim's interpretation. ASRF's design (boundary regression branch + smoothing) is exactly aimed at over-segmentation, and the paper reports its largest relative gains on edit/F1 (the over-segmentation-sensitive metrics) rather than frame accuracy ? so the interpretation is well-grounded, not an overreach.

7. user (2026-06-16T03:24:49.857Z)

Structured output provided successfully

8. assistant (2026-06-16T03:24:52.328Z)

Verification complete. The claim is ****not refuted**** ? it is an exact, faithful quote from the primary peer-reviewed source (ASRF, WACV 2021), corroborated by three independent sources, with the "up to" qualifier correctly preserved and the metric framing accurate.